



PROJECT CHARTER

1. **Project Title:** HealthPulse AI-Powered Patient Flow & Predictive No-Show Intelligence
2. **Project Sponsor:** (HealthPulse Clinic)
3. **Project Manager:** (Group 10 PMs)
4. **Project Start Date:** (18th Feb, 2026)
5. **Project End Date:** (25th Feb, 2026)
6. **Project Purpose/Business Need:**
 - HealthPulse is experiencing challenges due to high no-show rates (22% vs. industry average of 12%), leading to disrupted workflows and an estimated \$1.2M in annual revenue loss.
 - Patient dissatisfaction due to long wait times (45 minutes average, target is 15 minutes).
 - Operational inefficiency due to uneven doctor utilization (some at 95% capacity, others at 40%).
 - Siloed data across clinics hinders effective decision-making and proactive management.
 - The project aims to address these challenges by leveraging AI to predict no-shows, optimize patient flow, and improve operational efficiency.
7. **Project Objectives (SMART):**
 - **Reduce** the clinic-wide no-show rate from 22% to 12% (industry benchmark) within one week.
 - **Decrease** average patient wait time from 45 minutes to 15 minutes within one week.
 - **Improve** provider utilization balance across all eight clinics to minimize burnout and underutilization within one week.
 - **Establish** a structured governance framework to support predictive patient flow management within one week.
 - **Develop** a clear implementation roadmap that enables the transition from reactive scheduling to proactive operational planning within one week.
 - **Strengthen** organizational alignment through defined roles, accountability, and oversight structures within one week.
8. **Project Scope:**
 - **In Scope:**
 - Development of a predictive no-show model using historical patient data.

- Development of an AI-powered patient flow management system.
- Creation of interactive dashboards to visualize key performance indicators (KPIs).
- Integration of the solution with existing HealthPulse systems.
- Establishment of a data warehouse and data pipelines.

• **Out of Scope:**

- Implementation of new hardware or infrastructure (beyond what's listed in the tech stack).
- Extensive customization of existing HealthPulse systems.
- Training for end-users (separate training program will be defined).

9. Key Deliverables:

• **Technical Package:**

- Trained no-show model
- FastAPI endpoint (<300ms response)
- Complete data warehouse schema
- Data dictionary

• **Analytics Package:**

- Interactive dashboard
- Utilization heatmaps
- At-risk patient lists

• **Project Package:**

- Final insights report
- 3-5 actionable recommendations
- User guide
- Stakeholder presentation

10. Project Team Roles and Responsibilities:

- **Project Manager:** Keep timeline, communicate with stakeholders, track risks.
- **Business Analyst:** Define requirements, validate success metrics, user testing.
- **Data Engineer:** Build data pipeline, create warehouse, ensure data quality.
- **Data Scientist:** Develop prediction model, build API, tune performance.

- **Data Analyst:** Create dashboards, generate insights, support managers.

11. Key Stakeholders:

- Dr. Elena Martinez (Founder)
- (Other key stakeholders to be identified - Clinic Managers, Providers, IT Department)

12. Project Budget:

- \$250,000 (one-time investment - from project brief)

13. Project Risks and Assumptions:

• Risks:

- Data quality issues may impact model accuracy.
- Integration with existing systems may be more complex than anticipated.
- Stakeholder resistance to change.
- Limited availability of key personnel.

• Assumptions:

- Historical patient data is accurate and complete.
- Existing systems are well-documented and accessible.
- Stakeholders are committed to the project and will actively participate.

14. Project Success Criteria:

- Achievement of the stated project objectives (no-show rate reduction, wait time reduction, improved provider utilization).
- Successful implementation of the AI-powered patient flow management system.
- Positive feedback from stakeholders.
- Delivery of all key deliverables within the agreed-upon timeframe and budget.

15. Approval:

- (Signatures of Project Sponsor and Project Manager)